Biol Invasions (2012) 14:1305–1310
DOI 10.1007/s10530-011-0157-x

INVASION NOTE

Recent invasion and status of the raccoon (*Procyon lotor*) in Spain

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Received: 7 April 2011 / Accepted: 10 December 2011 / Published online: 24 December 2011

Abstract In this study we present the results of a baseline study designed to assess the status of the raccoon (*Procyon lotor*) throughout Spain. The species was reported in 28 localities, mostly consisting of sporadic observations of single individuals. In central Spain an apparently thriving population of raccoons has been recently discovered. Our data confirmed the spread of feral raccoons throughout this region, where the species has already colonized about 100 km of streams and rivers. Predation on local fauna was also proved, and the first approximation for spatial movement and habitat use analyses in Spain is presented. Our results suggest that deliberate releases of raccoons by pet owners are an important cause for the existence of feral raccoons in Spain. Further research should focus on monitoring established individuals to collect detailed data on their population and reproductive parameters. Meanwhile, urgent actions should be taken to stop releases into the wild and to control and eradicate this unwelcome invasive species.

Keywords Raccoon · *Procyon lotor* · Spain · Invasive alien species

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Introduction

The Raccoon (*Procyon lotor*) is a medium-size carnivore whose native distribution extends from Southern Canada to Central America. The species can be found in a wide range of habitats, including open and marshy areas, rivers, and urban areas (Bartoszewicz 2006). Non-native populations of raccoons have been established around the world, including Alaska, the Antilles, Japan and Europe (Frantz et al. 2005; Schrader and Hennon 2005; Okabe and Agetsuma 2007; Helgen et al. 2008). After the first introduction in Germany in 1927, the species successfully spread into most European countries (reviewed in Bartoszewicz et al. 2008 and in Canova and Rossi 2008). In Sweden, Great Britain, Norway and Denmark the species has been observed sporadically as a result of escapes or releases from fur farms, animal parks, or pet owners (Kauhala 1996).

It has been suggested that the introduction of raccoons might have a negative impact on the native fauna, although empirical evidence is still scarce

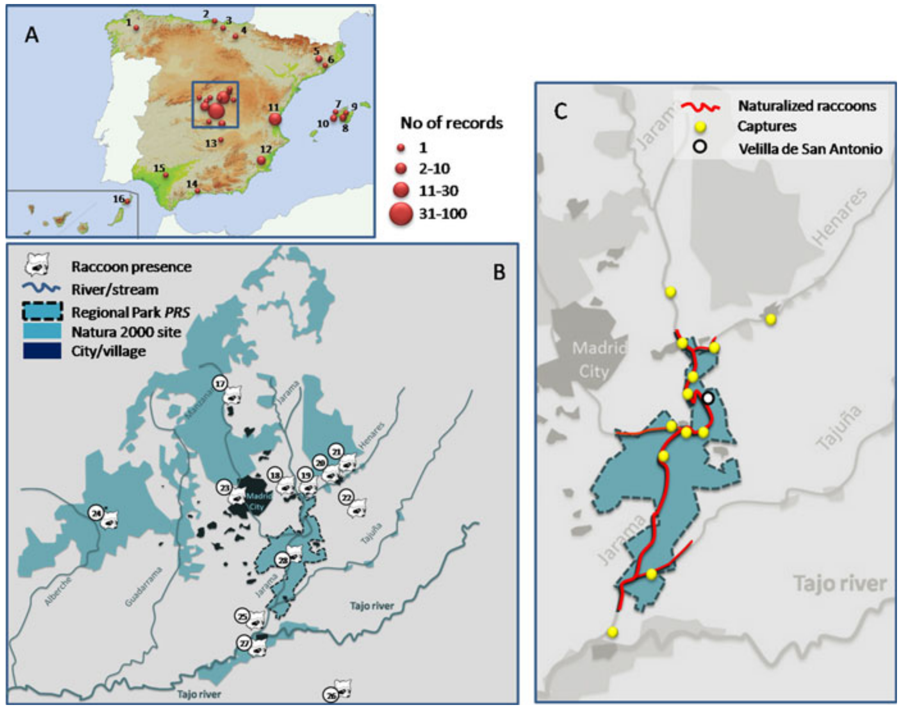

Fig. 1 **a** Map showing records of raccoon in Spain. Size of *dots* is proportional to the number of individuals reported. *Numbers* correspond to ID of localities in Table 1. **b** Map of the province of Madrid (central Spain) indicating the location of the Regional Park (*PRS*) and the localities where raccoons have been found (codes as in Table 1). **c** *Inset* showing in detail the limits of the *PRS* and the areas where wild populations of raccoon are known to occur. The *open dot* indicates the locality of Velilla de San Antonio, where the first raccoon was reported in this region (see "Results" and "Discussion")

(e.g. Kauhala 1996; Frantz et al. 2005). Raccoons are agile predators, feeding on a variety of molluscs, fish, amphibians, insects, birds, bird eggs, small mammals, vegetables and fruits, and also carrion and garbage (Bartoszewicz et al. 2008). Furthermore, there is growing evidence about the potentially negative impact of raccoons as disease vectors on the native fauna, domestic animals and even humans (e.g. Wandeler and Salsberg 1999). Consequently, the potential impact of this alien species goes beyond a detrimental relationship for the native fauna via predation or competition, and might be considered both as an ecological and a health risk.

This paper is the first report of the existence of raccoons in Spain. Our main objectives are to assess the current status of this exotic species in Spain, to explore the probability for successful reproduction in the wild, and to provide preliminary data on diet and movements in Spain.

Materials and methods

Raccoon distribution in Spain

Data on raccoon presence were obtained from field observations, road-kill records, trapping sessions, and indirect data collected by staff of the wildlife services from regional and national administrations from 2001 to 2010 (Fig. 1a; Table 1). Grey literature regarding raccoon occurrence in Spain was also investigated by searching the Spanish National and Regional Park Databases (annual reports), and reviewing the content of Spanish non-scientific journals during the 2000–2010 period. All records were GPS-positioned and incorporated to a Geographic Information System (Arcview 3.2 software).

Study area in central Spain

The central region of Spain is dominated by typical Mediterranean climate and vegetation, with a well-developed drainage network (Fig. 1b). The Tajo river drains the central Spanish Plateau and flows from east to west into the Atlantic Ocean in Lisbon. It is the longest river in the Iberian Peninsula (1,200 km) and the third largest in catchment area (81,947 km^2). This study was principally conducted in the Regional Park *Parque Regional del Sureste* (*PRS* hereafter), a Natura 2000 site located in the northeastern limit of the Tajo basin (Fig. 1b). Floodplains comprise the majority of this area, characterized by marshy meadows, irrigated crops (*Zea mays*, *Triticum* spp.) and wooded areas along the riverbanks (broad-leaved species, such as

Table 1 Records of raccoons in Spain

ID	Locality	Region	Year/period	Type of data	No. of animals	References
1	Becerreá	Lugo	2009	TA (loc)	1	
2	Santoña	Cantabria	2007	PHT (exp)	1	
3	Encartaciones	Vizcaya	2008	FD (loc)	1	
4	Bujanda	Álava	2010	CAP (exp)	1	
5	Begues	Barcelona	2001	CAP (exp)	3	
6	Mataró	Barcelona	2010	CAP (exp)	1	
7	Banyalbufar	Mallorca	2009	PHT (exp)	1	
8	Algaida	Mallorca	2001–2007	CAP, OBS (exp, loc)	3	
9	Lloret de Vistalegre	Mallorca	2006–2007	PHT (exp)	1	Pinya et al. (2009)
10	Puigpunyent	Mallorca	2006–2007	OBS (loc)	2	Pinya et al. (2009)
11	Valencia	Valencia	2003–2009	CAP (exp)	28	Technical report, unpublished
12	Murcia city	Murcia	2006–2008	CAP (loc)	3	
13	Calzada de Calatrava	Ciudad Real	2010	CAP (exp)	1	
14	Málaga city	Málaga	2010	CAP (exp)	1	
15	Sevilla city	Sevilla	2009	CAP (loc)	1	
16	Timanfaya	Lanzarote	2003	CAP (exp)	1	Technical report, unpublished
17	Soto del real	Madrid	2007	TA (loc)	1	
18	Torrejón de Ardoz	Madrid	2008	CAP (exp)	1	
19	Alcalá de Henares	Madrid	2010	PHT (exp)	1	
20	Azuqueca de Henares	Guadalajara	2009–2010	CAP (aut)	18	García et al. (2009)
21	Guadalajara city	Guadalajara	2010	CAP (pol)	1	
22	Escariche	Guadalajara	2009	TA (loc)	1	
23	Casa de campo	Madrid	2008–2009	OBS (loc)	3	
24	Chapineria	Madrid	2008	TA (loc)	1	
25	Seseña	Toledo	2009	CAP (exp)	2	
26	Miguel Esteban	Toledo	2009	CAP (loc)	1	
27	Aranjuez	Madrid	2009–2010	OBS (aut, loc)	2	
28	*PRS*	Madrid	2003–2010	CAP (aut)	78	Technical report, unpublished

Codes (ID) for localities as in Fig. 1

CAP captured animal, *PHT* photo-trapping, *TA* traffic accident, *FD* found dead, *OBS* observation (*aut* authors, *exp* experts from National or Regional Agencies, *loc* local people, *pol* police service)

Salix spp., *Populus* spp., etc.). The *PRS* is ca. 20 km south of the city of Madrid, and the landscape in between is mostly urbanized and densely populated.

Raccoon trapping at PRS

Trapping sessions were carried out in *PRS* during 2007–2010 in order to remove individuals from the wild, verify their parasitological status, and study their diet. Trapping was conducted in seven riparian sites throughout the *PRS* (Fig. 1c) using home-made collapsible wire-boxes constructed of galvanized wire mesh (120 × 30 × 35 cm). Traps were set on major animal trails in all locations with previous raccoon sightings and wherever signs of raccoon activity were found (tracks, latrines, etc.). All traps were checked daily and baited with commercial peanut butter in combination with visual attractants (two fresh chicken eggs) every 3 days. If no signs of animal activity were noticed after 2 weeks, traps were removed and relocated. Thus, trapping periods in each site lasted between 2 and 8 weeks depending on capture success. All individuals captured were visually inspected to determine age and sex, and euthanized by veterinaries from the regional administration (Spanish Wildlife Recovery Centres *CRAS*). Two adult males captured during trapping sessions were fitted with radio transmitters (VHF Tellus TXV-10, Telonics Inc.) and released after vasectomization, with the purpose of finding new individuals (Judas technique). Animals were located with a portable receiver (Yaesu, Japan) and a 3-element Yagi antenna. Two diurnal locations were taken daily during the summer (June–August), and 2–4 locations per week during the rest of the year. Additionally, the information retrieved from their fixes was analyzed in order to get preliminary data on dispersal movements and habitat use of the raccoon in Spain. Home range size and shape were determined using the 95% Minimum Convex Polygon (MCP95).

A preliminary exploration of the raccoon's diet in the study area was made using stomach content samples (n = 10) from sacrificed raccoons. Also, during weekly surveys to detect raccoon presence in *PRS*, scats found in latrines on tree forks (n = 51) and remains of animals predated by raccoons (n = 18), were collected and included in the diet analyses. Scats were determined as belonging to raccoons according to their size, shape, smell and location, as none of other co-occurring mammal species in the study area use latrines on tree forks. Prey remains consisted of headless carcasses located on top of riverbank trees, often with clearly visible footprints at the base of the tree. There is no evidence of such behaviour in any native mammal or avian predator in the area. Prey species were identified according to bone remains and hair or feather characteristics. Diet composition was expressed as the percentage of occurrence of each prey category in the total sample (i.e. number of samples with presence of a particular prey compared with the total number of samples).

Results

Raccoon presence in Spain

We collected data from 160 records in 28 localities (considering the population of *PRS* as one locality) (Table 1; Fig. 1a, b). Records were concentrated in central Spain, mainly within the limits of the *PRS* and neighbouring sites. The second largest population occurs in Guadalajara, located 78 km north of the *PRS* and 60 km north of Madrid. The northernmost record in continental Spain was found in Santoña (Cantabria province) and the southernmost in Málaga (Fig. 1a; Table 1). In the Balearic Islands, presence of raccoons was reported in the island of Mallorca (Pinya et al. 2009) in at least four different sites (Fig. 1; Table 1). In the Canary Islands, one adult raccoon was captured in 2003 in the Timanfaya National Park (Lanzarote) after predating over one hundred Cory's shearwaters (*Calonectris diomedea*) in a protected breeding colony. Most records across Spain consist of sporadic observations of single individuals without apparent connection to the largest population in central Spain.

Table 2 Raccoon trapping data at Regional Park *Parque Regional del Sureste* (Madrid, Spain)

Year	Number of traps-night	Captures per 100 traps-night	Captures		
			Adults	Juveniles	Cubs
2007	320	4.7	1	6	8
2008	555	0.9	5	0	0
2009	1,384	1.87	14	8	4
2010	2,612	1.18	15	16	0

Trapping sessions at PRS

Trapping effort and success differed among years (Table 2). A total of 77 individuals were captured at *PRS* during the study period. Necropsies showed nine individuals (12%) with clear signs of breeding in the wild (e.g. lactating females and cubs born in the same year of capture). In 2008 a well-established breeding population was considered to exist along 20 km of the Jarama, Henares and Tajuña riverbanks (García et al. 2007). Since then, the species has been reported in 12 different localities in that area (Table 1; Fig. 1b, c). Currently, raccoons occupy over 28 km of riverside habitat (Fig. 1c), reaching very high densities in some river stretches (e.g. five captures in one night using 11 traps situated along 200 m of riverside).

Food intake (stomach contents) mainly consisted of red swamp crayfish (*Procambarus clarkii*) and feathers of aquatic birds, present in 60 and 50%, respectively, of the stomachs analyzed (n = 10). Faecal analyses showed bones and hair of wild rabbit (*Oryctolagus cuniculus*) and wood mouse (*Apodemus sylvaticus*) in 15% of scats analyzed (n = 51), and fruit remains and seeds (grape, fig and blackberry) in 100% of the samples. The 18 carcasses found predated by raccoons corresponded to *Anas strepera* (n = 12, 67%), *Anas platyrrhynchos* (n = 4, 22%), unidentified *Anas* spp. (n = 1, 5%) and *Gallinula chloropus* (n = 1, 5%).

For the two radio-tagged animals, we obtained 72 and 35 fixes, during 12 months (November 2009 to October 2010) and 6 months (April to October 2010) of survey, respectively. The maximum distance between fixes was 5.5 km for the former and 10.2 km for the latter individual. None of the animals were found more than 1 km away from any river or stream (896 and 596 m, respectively). Home range sizes were 410 and 1,180 ha, as estimated by the 95% MCP.

Discussion

Since the first release in Germany in 1927, the raccoon has spread throughout Europe in a relatively short time. In Spain it has recently been reported as an exotic species (first records in 2001) and it is already widely distributed (Fig. 1a). A large number of adult raccoons and cubs were found near Madrid (Barona and García-Román 2005), Barcelona and Valencia, which could be indicative of extensive releases from pet owners in such densely populated cities. Raccoon sightings were most frequent in central Spain, where the species successfully reproduces and has already colonized about 100 km of riverbanks (Fig. 1b). Its distribution range in that region includes the original locality where the species was sighted for the first time (Velilla de San Antonio, Madrid), the Henares river to the north (up to the city of Guadalajara) and the Jarama river to the south (to the city of Aranjuez, ca. 50 km south from Madrid) close to the Tajo river. According to capture records and radio-tagging data, raccoons at *PRS* occurred mainly close to rivers, which seem to be acting as corridors for raccoon dispersal outside the *PRS* (both northwards and southwards). The expansion and proximity to the Tajo river poses a serious problem for the control of the species, because this major river, although it may represent a barrier to the south, it could also facilitate raccoon dispersal to the west of Spain and Portugal.

In addition to the potential threats in Atlantic and Mediterranean Spanish islands, which hold a high rate of endemic species, Central Spain is considered an important biodiversity hotspot within the Iberian Mediterranean landscapes, encompassing a wide variety of habitats that sustain an important diversity of faunal and floral species. The biological importance of the Iberian Peninsula is not limited to the richness or uniqueness of its resident fauna and flora. It constitutes an important stopover region for millions of migratory birds (Martí and Del Moral 2002), including seven areas classified as Special Protection Areas for Birds in central Spain. Thus, the presence of the raccoon in such protected areas, considering the adaptable nature of its diet, might lead to catastrophic consequences for the native fauna. There are no precise quantitative indicators regarding this issue, but our preliminary results from diet analyses, together with previous experiences (e. g. predation on a breeding colony of *Calonectris diomedea* described above), lead us to suspect that the raccoon in this region might have a strong impact against the local fauna. Raccoon predation on ground nesting birds has been commonly reported in the literature (examples in Zeveloff 2002; Ikeda et al. 2004; Bartoszewicz et al. 2008), as well predation on amphibians and reptiles (Kauhala 1996) and on clutches of the Spanish terrapin *Mauremys leprosa* (Álvarez 2008). Indeed, the existence in central Spain of large water sources where hundreds of ducks are temporarily flightless when moulting, could favour predation by raccoons, and would likely explain the amount of predated waterfowl carcasses found during this study. In addition, there is extensive evidence for the species potential to act as a reservoir for a broad array of animal and human pathogens (Rabinowitz and Potgieter 1984; Riley et al. 1998; Randall et al. 2007), including several documented cases of raccoon roundworm (*Baylisascaris procyonis*) infections in United States and Canada (Park et al. 2000), and at least one case in Europe (Küchle et al. 1993).

Considering the raccoon as an undesirable alien species that should be eradicated, the rapid population growth evidenced in this study suggests the need for a trans-regional monitoring plan and integrated actions before the population runs out of control (Mack et al. 2000; Simberloff 2003). Research and funding are urgently needed to determine its current abundance, distribution, diet and parasitological status. Therefore, the pet trade must be more strictly regulated to stop the release of pets into the wild. Furthermore, control and eradication programs should necessarily include efforts for public awareness against invasive alien species, especially when they imply irreversible impacts on native ecosystems and human sanitary risks. Finally, we propose the development of official campaigns to retrieve potentially invasive pet species as a primary step to face this problem.

Acknowledgments We would like to thank the staff working on wildlife conservation in the Regional Park *Parque Regional del Sureste*, Comunidad Autónoma de Madrid, Comunidad Autónoma de Valencia (J. Jiménez) and Comunidad Autónoma de Castilla-La Mancha (Guadalajara Province) for their help in this research (captures) and for providing data on raccoon presence. Capture authorizations were provided by the Comunidad Autónoma de Madrid according to the EU laws. The study was partially financed by the Regional Park *Parque Regional del Sureste* and the Comunidad Autónoma de Madrid.

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